REPEATING DECIMALS · MULTIPLICATIVE ORDER · CYCLE TYPES

# Reptend primes

$c \mid p-1$

The repeating block of $1/p$ has full length when long division visits every nonzero remainder.

1 / FOLLOW THE REMAINDERS

## The first return to $1$ gives the decimal period

For a prime $p \neq 2, 5$, start long division of $1$ by $p$ at $r_0 = 1$. At each step, $10r_j = p\,d_{j+1} + r_{j+1}$, where $d_{j+1}$ is the next digit and $r_{j+1}$ the new remainder. The tail of the decimal at step $j$ is $r_j/p$, so a return to $r_0$ repeats the whole tail.

$$\underbrace{1 \xrightarrow{1} 3 \xrightarrow{4} 2 \xrightarrow{2} 6 \xrightarrow{8} 4 \xrightarrow{5} 5 \xrightarrow{7} 1}_{\text{remainders for } p=7;\ \text{arrow labels are digits}} \qquad \frac{1}{7} = 0.\overline{142857}, \quad c = 6.$$

The remainders satisfy $r_j \equiv 10^j \pmod p$. So the shortest period $c$ is the least positive $k$ with $10^k \equiv 1 \pmod p$. This is the **multiplicative order** of $10$ modulo $p$.

2 / USE THE TOTIENT

## Why the period divides $p-1$

Since $\varphi(p) = p-1$ and $\gcd(10, p) = 1$, Euler's theorem gives $10^{p-1} \equiv 1 \pmod p$. Write $p-1 = qc+s$ with $0 \le s < c$. Use $10^c \equiv 1$ to remove the $q$ full blocks:

$$10^{p-1} = (10^c)^q 10^s \equiv 10^s \equiv 1 \pmod p.$$

The first positive return is at $c$, so $s = 0$ and $c \mid p-1$. The same argument works for any return exponent.

**Full reptend.** If $c = p-1$, the decimal uses every nonzero remainder in one cycle. Then $10$ is a **primitive root** modulo $p$.

3 / START FROM A DIFFERENT REMAINDER

## The cycle type counts whole cycles

Starting at a nonzero remainder $a$, return means $10^k a \equiv a \pmod p$. Cancel $a$: the first return is still at $k = c$. Multiplying by $10$ is reversible, so all $p-1$ nonzero remainders split into cycles of length $c$. Their number is $m = (p-1)/c$.

**Two cycles for $13$.** Each list loops back to its first entry:

$$(1, 10, 9, 12, 3, 4), \qquad (2, 7, 5, 11, 6, 8).$$

So $1/13 = 0.\overline{076923}$ and $2/13 = 0.\overline{153846}$. When $m = 1$, all numerators $1, \ldots, p-1$ start on the same cycle. Their digit blocks are rotations, as in $2/7 = 0.\overline{285714}$.

4 / READ THE LINES IN THE DATA

## Each cycle count gives a line of slope $1/m$

Plot $c$ against $p-1$. Each point lies on the line

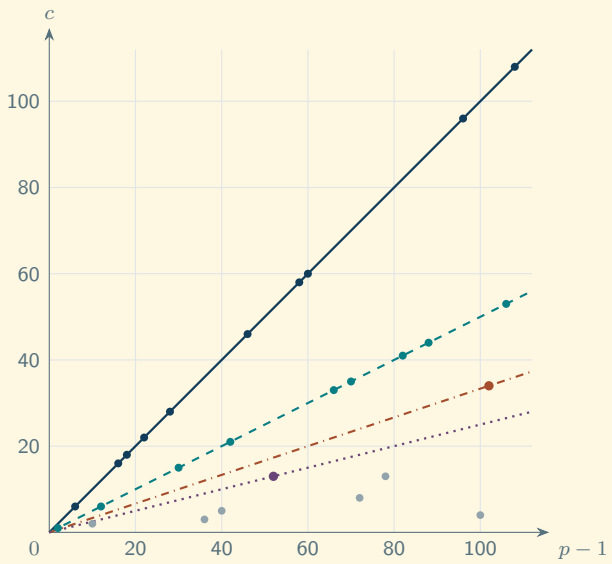
$$c = \frac{p-1}{m}.$$

The full reptend primes sit on the top line, $m = 1$.

| Type $m$ | Prime $p$ | Period $c$ |
|---|---|---|
| 1 | 7 | 6 |
| 2 | 43 | 21 |
| 3 | 103 | 34 |
| 4 | 53 | 13 |

**Check $p = 43$.** Modulo $43$, $10^{21} \equiv 1$, $10^3 \equiv 11$ and $10^7 \equiv 6$. Every proper divisor of $21$ divides $3$ or $7$, so $c = 21$ and $m = 42/21 = 2$. Type $4$ is impossible there because $4$ does not divide $42$.

**Count the dots.** In this range, $10$ of the $27$ eligible primes are full reptend. The lines classify periods; the dots give the counts.




Dots: all eligible primes up to $109$. Colours match the table; grey dots have $m > 4$.

**Try $p = 17$.** Given $10^8 \equiv -1 \pmod{17}$, use $c \mid 16$ to find $c$ and $m$.

**Check:** every proper divisor of $16$ divides $8$. Since $10^8 \not\equiv 1$, the period is $16$ and $m = 1$.

For $p = 2, 5$, the decimal terminates. In base $b$, use $b$ in place of $10$ and assume $p$ does not divide $b$.

Based on *Reptend Primes Interactive Explorer*. Reference: Pieter Moree, *Artin's Primitive Root Conjecture*, §1.

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